Physics - Graduate Level Assessment 5 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. A water pipe in which water pressure is 80 lb/in.^2 springs a leak, the size of which is .01 sq in. in area. How much force is required of a patch to "stop" the leak?
   (a) 8 oz
   (b) 0.008 lb
   (c) 8 kg
   (d) 800 lb
   (e) .8 lb

2. Anucleusof mass M emits a Υ ray of energy E_Υ^xnc . The nucleuswas initially at rest . What is
   (a) E_^2/ 4Mc^2
   (b) $2E_\Upsilon^2/Mc^2$
   (c) E_Υ/Mc^2
   (d) $3E_\Upsilon^2/2Mc^2$
   (e) $E_\Upsilon/4Mc^2$

3. When 100 J are put into a device that puts out 40 J of useful work, the efficiency of the device is
   (a) 25%
   (b) 100%
   (c) 60%
   (d) 50%
   (e) 75%

4. Find the minimum energy of a simple harmonic oscilla-tor using the uncertainty principle.
   (a) $\frac{h\omega}{4\pi}$
   (b) mω^2/2
   (c) $\frac{h}{m\omega^2}$
   (d) $\frac{h^2}{2\pi m\omega}$
   (e) h^2/mω

5. What is the total charge in coulombs of $75.0 ~kg$ of electrons?
   (a) -2.05×10^{13}C

(b) $-1.32\ 10^{13}$C

(c) $-1.98\ 10^{13}$C

(d) -0.98×10^{13}C

(e) -1.44×10^{13}C

True / False (10 marks)

Answer True or False

6. True or False: An electron can reach a speed of 1.8 x 10^7 m/s by being accelerated across a po
7. True or False: A uniformly charged conducting sphere with a diameter of 1.2 m and surface charg
8. True or False: An atom with filled n = 1 and n = 2 energy levels contains a total of 12 electrons.
9. True or False: The focal length of the lens is -20 inches, indicating that it is a diverging lens.
10. True or False: The specific heat capacity of sand is always equal to that of vegetation, leading to

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Analyze the formation of the image by a bi-convex lens with a radius of curvature of 20 cm when an object of height 2 cm is placed 30 cm away. Discuss the characteristics of the image produced.
12. An observer moves past a meter stick at a velocity of half the speed of light. Discuss how relativistic effects influence the measured length of the meter stick and calculate the observed length.

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